AI-Native Brand Identity: From Visual Recognition to Cryptographic Verification

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Abstract

This paper advances four contributions to brand identity research. First, we propose the *observer-driven evolution thesis*: identity verification technologies change discontinuously in response to shifts in the observer type. Wax seals served bureaucratic observers, hallmarks served asymmetric-information buyers, trademarks served mass-market consumers, and SSL certificates served web browsers. As AI agents increasingly mediate purchasing decisions and brand evaluation, this pattern predicts a structurally different identity technology. Second, we introduce *behavioral metamerism*: the phenomenon where brands produce identical statistical profiles but exhibit structurally different behavioral patterns, creating AI-native brand confusion that statistical optimization cannot resolve. We extend Spectral Brand Theory to define the Brand Function — f(query, context, observer_type, time) -> response — as the complete behavioral specification of how a brand operates when observed by machines, and argue that cryptographic signatures on behavioral specifications will serve as the primary brand identity mechanism for AI-mediated commerce. Drawing on emerging standards (W3C Verifiable Credentials, GLEIF vLEI, .well-known/brand.json) and empirical evidence on LLM brand bias, we propose six formal propositions connecting identity verification technology to observer type, behavioral specification to brand coherence, and cryptographic attestation to trust in autonomous commerce. We formalize behavioral metamerism as an observational equivalence condition and propose a pilot study design. Logos will persist for human observers but face structural irrelevance for AI agents; organizations face pressure to develop dual identity infrastructure: visual for humans, cryptographic for machines.

Keywords: brand identity, AI agents, cryptographic verification, behavioral specification, metamerism, agentic commerce, generative engine optimization, observational equivalence

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1. Introduction

Every time the observer of brand identity changed, the verification technology changed with it. Wax seals authenticated documents for literate officials. Hallmarks guaranteed metal purity for buyers who could not assay it themselves. Trademarks enabled recognition in anonymous mass markets. SSL certificates proved server identity to web browsers. The logo — the dominant identity technology of the 20th century — was designed for a specific observer: the human consumer navigating scaled markets through visual and emotional pattern matching.

That observer is changing. AI purchasing agents — Amazon Rufus, Google AI Mode, Perplexity Shopping, ChatGPT with agentic commerce capabilities — are increasingly mediating product

discovery, brand evaluation, and purchasing decisions. These systems do not process logos, emotional narratives, or visual brand guidelines. They process structured data, behavioral patterns, and verified claims. A growing body of evidence documents how large language models (LLMs) form "brand perceptions" that are statistical rather than perceptual: global brands are favored due to training data frequency (Kamruzzaman et al., 2024), market-dominant products receive disproportionate recommendation (Zhi et al., 2025), and optimization strategies designed for human search engines produce diminishing returns in generative AI environments (Aggarwal et al., 2024). Current brand identity infrastructure — the visual brand book, the logo, the tagline — is built entirely for human observers and carries zero informational weight for machine observers.

Kannan and Li (2017) mapped the digital touchpoints where technology mediates the marketing process — from search and social to mobile commerce. The AI agent is the next generation of that digital mediation, but it differs structurally from all prior digital touchpoints: it does not present information to a human who then decides, but evaluates and decides itself. The Brand Function extends the touchpoint framework Kannan and Li proposed to this agentic era, where the touchpoints are machine-to-machine and the brand must specify behavior for an observer that processes structured data rather than sensory experience.

The dominant paradigm in brand identity theory — Keller's (1993) customer-based brand equity model, which locates brand equity in the knowledge structures of human consumers — is built entirely for human observers. Keller's four dimensions of brand knowledge (brand awareness, brand associations, perceived quality, and brand loyalty) presuppose a human mind forming associative networks through direct and mediated experience. This human-observer assumption pervades the subsequent identity literature: Hatch and Schultz (2010) model brand co-creation as stakeholder dialogue — a process that requires observers capable of dialogue; de Chernatony (1999) frames brand identity as the gap between management vision and consumer perception — a formulation that loses meaning when the "consumer" is a recommendation algorithm with no perception in the phenomenological sense. Hallinan and Striphas (2017) identify how algorithmic culture reshapes what content reaches consumers, but do not address how brands should present identity *to* the algorithms themselves. When the observer is an AI agent forming "associations" through statistical co-occurrence in training data, the Keller framework describes a process that does not apply, the co-creation model has no counterpart, and the vision-perception gap becomes a specification-implementation gap.

This paper identifies a gap at the intersection of three developing literatures. First, multiple protocols now address machine-readable brand identity (.well-known/brand.json, the Agentic Commerce Protocol, the Unified Commerce Protocol, Brando Schema) and cryptographic attestation (BIMI/VMC, GLEIF vLEI, GS1 Verifiable Credentials). Second, marketing scholarship has begun addressing algorithmic brand equity (Kovalenko et al., 2026), algorithmic branding through platform assemblages (Kozinets, 2022), and the platformization of brand management (Wichmann et al., 2022). Third, the agentic commerce literature has identified the "Shopper Schism" — the structural separation of the consumer (who decides what to want) from the shopper (who executes the purchase), with AI agents increasingly occupying the shopper role (Accornero, 2026). Puntoni et al. (2021) identify delegation as one of four core tensions in consumer-AI interaction — the

moment when a consumer transfers purchasing authority to an AI agent. The Brand Function addresses this delegation tension directly: when consumers delegate to AI agents, the behavioral contract provided by the Brand Function becomes the mechanism through which delegated purchasing decisions can be verified against the brand's intended identity. However, no theoretical framework connects these developments by explaining *why* these technologies are emerging now, *what* is missing from them, or *how* they fit into the longer historical pattern of identity technology evolution.

This paper makes four contributions. First, we propose the *observer-driven evolution thesis* — a synthesizing framework explaining why identity technologies change discontinuously when the observer type changes. While individual histories of seals, hallmarks, trademarks, and digital certificates exist in separate literatures (Richardson, 2008; Sáiz, 2018; Economides, 1988; Groebner, 2007), no prior work theorizes this as a general pattern or uses it to predict the next identity technology. Second, we introduce the *Brand Function* as a root specification concept — extending existing protocols, which address brand discovery and catalog, to include behavioral specification, perception measurement, and coherence verification. Existing protocols such as brand.json and Brando Schema address creative and communication governance; the Brand Function adds behavioral specification across all touchpoints, spectral perception dimensions from Spectral Brand Theory (Zharnikov, 2026a), and computable coherence metrics. Third, we introduce *behavioral metamerism* — the phenomenon where brands produce identical statistical profiles but exhibit structurally different behavioral patterns — as the AI-native form of brand confusion that statistical optimization cannot resolve. Fourth, we formalize these contributions in six propositions connecting observer type to identity technology, specification to coherence, and cryptographic attestation to trust in autonomous commerce.

The remainder of the paper proceeds as follows. Section 2 traces the historical pattern of observer-driven identity technology change across five transitions. Section 3 examines how AI agents currently perceive brands and why current identity infrastructure fails them. Section 4 introduces the Brand Function as a root specification. Section 5 develops the cryptographic signature as the AI-native equivalent of the logo. Section 6 introduces behavioral metamerism. Section 7 establishes the necessity conditions under which the Brand Function becomes structurally required. Section 8 discusses implications for brand management and marketing theory. Section 9 addresses limitations and future research directions. Section 10 concludes.

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2. Historical Pattern: Observer Determines Technology

Identity verification technologies do not evolve incrementally. They emerge in response to a change in the observer type. Five transitions establish the pattern that predicts the sixth. The examples trace the Western European genealogy that produced modern trademark law and digital identity infrastructure — the direct ancestors of the AI commerce protocols under analysis.

Wax seals served the first non-personal observer: the bureaucratic official verifying documents at a distance. Mesopotamian cylinder seals (ca. 3500 BCE) authenticated clay tablets through unique physical impressions. Medieval signet rings were destroyed upon death — an early form of key

revocation. The technology was adapted to an observer who could inspect a physical impression but lacked personal knowledge of the sender.

Hallmarks emerged when the observer shifted from official to commercial buyer. England's 1300 statute required silver to be assayed at Goldsmiths' Hall before sale, introducing trusted third-party attestation. The buyer needed only to recognize an institutional mark, not possess metallurgical expertise. Wax seals persisted for legal documents but were irrelevant for commercial quality verification.

Trademarks responded to the Industrial Revolution's anonymous consumer. Pre-industrial craft marks identified known local makers (Richardson, 2008); mass production broke this link. Trademarks carried abstract associations — quality, emotion, culture — rather than certifying physical properties (Schechter, 1925; Economides, 1988). Aaker (1996) and Kapferer (2008) codified this mature form into brand identity systems designed for a specific observer: the human consumer forming associations through sensory experience. The logo is the mature expression of this technology.

Holograms are instructive because the observer did *not* change. When industrialized counterfeiting intensified the threat but the observer remained human, the response was incremental — adding optical verification features to the existing visual paradigm. Discontinuous technology change required a change in the observer itself.

SSL certificates (Netscape, 1994) marked the decisive break. For the first time, the primary identity observer was a software program — the web browser. Verification became mathematical rather than perceptual, binary rather than graded, and scalable without human attention. Logos persisted on websites but carried zero weight in the browser's trust decision. This transition established the template: when the observer is a machine, the identity technology must be verifiable by computation, not perception.

Proposition 1: Identity verification technology characteristically changes not through incremental improvement but through discontinuous response to a change in the observer type.

Falsification: P1 is falsified if the current AI-observer transition produces only incremental extensions of existing logo-based identity (e.g., animated logos for AI agents, richer metadata on existing visual marks) rather than a structurally different verification mechanism. If logo-based identity proves sufficient for AI agents across multiple commerce categories, the discontinuity thesis fails.

Proposition 2: When the observer changes, the previous identity technology persists for legacy observers but becomes functionally irrelevant for the new observer class.

Falsification: P2 is falsified if AI agents demonstrably use logos as informational inputs in purchasing decisions — not merely recognizing them as identifiers but weighting visual brand identity in recommendation quality. If logo presence/absence significantly affects AI recommendation accuracy, the irrelevance claim fails.

The historical evidence supports both propositions. Wax seals persist in legal ceremonies; hallmarks persist in precious metals regulation; logos persist on websites despite SSL certificates handling actual identity verification. Each technology remains functional for its original observer class while becoming irrelevant for the new one. The question for marketing theory is: what happens when the observer changes again?

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3. The AI Agent as Observer

The sixth observer type is the AI agent. Unlike the web browser, which verified identity as a binary trust decision before displaying content for human evaluation, AI agents increasingly make substantive brand evaluations — comparing products, generating recommendations, executing purchases — with varying degrees of human oversight. In the taxonomy of Huang and Rust (2021), the AI purchasing agents examined here operate primarily at the "thinking AI" level — processing structured data to arrive at purchasing decisions — though emerging multimodal capabilities may introduce "feeling AI" components that the Brand Function's multi-dimensional specification anticipates. This section examines how AI agents currently perceive brands, what evidence exists for systematic bias in their evaluations, and why current optimization strategies are structurally insufficient.

3.1 How AI Agents Currently "See" Brands

AI purchasing agents process brand identity through structured data, behavioral signals, and third-party corroboration — none of which involves visual recognition. Current platforms illustrate the pattern:

Amazon Rufus draws primarily on Amazon's internal structured catalog, including product attributes, reviews, Q&A content, pricing competitiveness, and seller performance metrics. A brand's identity to Rufus is its aggregated review profile, fulfillment reliability, and catalog completeness. Industry analyses suggest that products below a four-star rating with fewer than approximately 9,000 reviews receive significantly reduced recommendation frequency.

Google AI Mode leverages the Shopping Graph, which integrates over 50 billion product listings with approximately 2 billion updates per hour from Merchant Center feeds, schema.org markup, and web data. Brand identity is a function of structured attribute completeness, pricing consistency, review quality, and freshness of availability data.

Perplexity Shopping emphasizes citation-driven evaluation, drawing on third-party earned media, authoritative web sources, and pricing consistency. A brand that lacks authoritative third-party coverage is effectively invisible regardless of its visual identity strength.

ChatGPT with agentic commerce uses the Agentic Commerce Protocol (ACP) for checkout and payment delegation, Stripe integration for transaction processing, and product feeds for discovery. Brand identity is whatever the structured data says it is.

Across all platforms, AI agents evaluate brand identity through data provenance, behavioral consistency, and structural completeness. The logo carries no informational weight. Comprehensive

schema.org markup can increase recommendation likelihood substantially, while visually striking brand identities with poor structured data are systematically disadvantaged.

3.2 LLM Brand Bias as Evidence of Statistical Perception

If AI agents formed brand perceptions through genuine understanding, their evaluations would reflect product quality independent of training data distribution. Empirical evidence contradicts this.

Kamruzzaman et al. (2024), in a systematic study published at EMNLP, demonstrated that LLMs exhibit strong brand bias correlated with training data frequency: global brands are recommended disproportionately over local alternatives, luxury brands are recommended 88–100% of the time for high-income country queries, and geographic and economic skew systematically distorts recommendations. The mechanism is statistical: brands that appear more frequently in training corpora receive higher probability mass in generative outputs, independent of objective quality.

Zhi et al. (2025) extended this finding to investment recommendations, showing that market-dominant products receive amplified recommendation due to neural reinforcement — a feedback loop where training data frequency creates recommendation bias, which generates more user interaction data, which further amplifies the bias. The effect is structural: it operates through the statistical properties of the model's learned representations, not through any process analogous to human brand evaluation.

Complementary evidence comes from the agent evaluation literature. Lyu et al. (2025) benchmark AI shopping agents on task completion — whether the agent can find the right product given multi-attribute queries with filters and sorting preferences across five product categories. Even the best deep research systems achieve only 30% holistic task success, and performance drops sharply with query complexity. This establishes that current AI agents fail at the navigation level before perception is even at issue. DeepShop evaluates agent *capability* (can it complete the task?) but not agent *perception* (does it perceive the brand correctly?). The present paper addresses the perception gap: the Brand Function provides the specification layer that ensures that when AI agents do successfully locate products, they also correctly represent the brand's multi-dimensional identity to the consumer. For empirical measurement of the perceptual failure mode, see Zharnikov (2026v).

These findings establish that LLM "brand perception" is fundamentally different from human brand perception. Humans form brand convictions through multi-sensory experience, social influence, personal history, and emotional resonance — processes that SBT models as observer-specific spectral weighting across eight dimensions (Zharnikov, 2026a). LLMs form "brand representations" through statistical co-occurrence patterns in text corpora. This distinction is consequential: it means that the identity infrastructure humans use (visual consistency, emotional narrative, brand experience) does not transfer to AI observers. De Bruyn et al. (2020), in a foundational JIM analysis of AI in marketing, identified biased AI and badly defined objective functions as structural pitfalls of AI marketing systems — behavioral metamerism is a direct consequence of both, arising when brands optimize for the same statistical objective (GEO visibility) and the AI's learned representations inherit training-data frequency bias. The broader JIM conversation

on AI-mediated marketing has established both the promise and structural risks: Ramesh and Chawla (2022) survey the chatbot marketing literature and identify brand personality transfer as a critical gap — chatbots can simulate brand voice but cannot verify brand identity, precisely the verification problem this paper formalizes. Pagani and Wind (2025) argue that AI unlocks marketing creativity but requires new governance frameworks to manage the loss of human oversight — a governance challenge that the Brand Function specification (Section 5) directly addresses. Shankar and Balasubramanian (2009) anticipated many of these dynamics in the mobile marketing context, noting that as marketing channels shift from human-centric to technology-mediated, identity verification mechanisms must evolve correspondingly.

3.3 Generative Engine Optimization as Symptom

Generative Engine Optimization (GEO) — the practice of optimizing content for visibility in AI-generated responses rather than traditional search rankings — represents the first systematic attempt by brands to be "seen" by AI observers. Aggarwal et al. (2024) introduced the GEO framework at KDD, demonstrating that strategies such as citation addition, statistics incorporation, and authoritative source references can increase brand visibility in generative AI outputs by up to 40%. Acar and Schweidel (2026) document cases in practitioner research where LLMs miscategorize brand positioning — a prestige Scotch brand (Ballantine's) described as a mass-market offering — providing real-world evidence for the systematic AI misperception that the present paper theorizes as behavioral metamerism.

However, GEO addresses a symptom rather than the underlying structural problem. GEO optimizes for *statistical visibility* — increasing the probability that a brand appears in generative outputs. It does not address *identity verification* — establishing that the brand is who it claims to be and behaves as it specifies. This distinction is critical. A brand can achieve high GEO scores through content optimization while having no machine-verifiable identity, no behavioral specification, and no way for an AI agent to verify that the entity it encounters is the entity it represents itself to be.

Furthermore, GEO creates a convergence dynamic. As brands in the same category adopt similar GEO strategies — adding citations, incorporating statistics, optimizing for the same query patterns — their statistical profiles converge. This convergence is precisely the condition that produces behavioral metamerism, which Section 6 develops in detail.

Proposition 3: AI agents do not perceive brand identity through visual recognition but through structured data evaluation, behavioral pattern analysis, and third-party corroboration. The logo carries near-zero informational weight for text-based AI agents, and minimal informational weight in text-only AI mediation more broadly.

Caveat on multimodal agents: Multimodal AI agents (GPT-4V, Gemini Vision) do process visual elements including logos, packaging, and color. The near-zero weight claim applies specifically to text-based agents operating through structured data and language models. As multimodal agents gain commerce capabilities, the visual identity layer may recover partial informational weight — though the structural data and behavioral specification layers are predicted to remain primary.

Falsification: P3 is falsified if multimodal AI agents that process visual brand elements (logo, packaging, color) make measurably better purchasing decisions than text-only agents with identical structured data. If visual brand identity provides incremental information value to AI agents beyond what structured data conveys, the "near-zero weight" claim is too strong.

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4. The Brand Function

If AI agents perceive brands through behavioral patterns rather than visual marks, then the brand needs a behavioral specification — a machine-readable, verifiable description of how it operates. This section introduces the Brand Function as the root specification from which all brand emissions derive, examines the current protocol landscape, and identifies what existing specifications fail to address.

4.1 The Brand Function as Root Specification

In software engineering, the specification defines correct behavior before implementation begins. A system without a specification cannot be verified, tested, or audited — one can detect anomalies relative to past behavior but cannot determine whether the behavior conforms to the system's intended purpose (Dijkstra, 1970; Clarke et al., 1999). The same principle applies to brands.

We define the Brand Function as:

f(query, context, observer_type, time) -> response

Where *query* is what an observer (human or machine) is asking of the brand; *context* is the observer's state, including prior knowledge, optimization objectives, and interaction history; *observer_type* distinguishes among humans, AI purchasing agents, AI research agents, AI comparison agents, and other observer classes; and *response* is the structured output that carries the brand's behavioral signature.

The Brand Function is the root specification from which all brand emissions and identity artifacts derive. This framing makes a deliberate architectural claim: *technologies are the primary rendering engines of brand behavior; the organization is coordination overhead.* When brand behavior is rendered through an API response, a pricing algorithm, a chatbot interaction, or a smart contract, the technology implements the Brand Function directly. The organizational structure — the people, processes, and hierarchies that coordinate brand behavior — is a necessary but lossy intermediary that introduces what Spectral Brand Theory terms a "specification gap" between intended and actual behavior (Zharnikov, 2026a). This parallels Parasuraman et al.'s (1985) SERVQUAL Gap 2 (the difference between management perception of consumer expectations and service quality specifications), but extends it: the specification gap applies not only to service delivery but to all brand emissions, and it is computable rather than subjective.

As AI agents increasingly render brand behavior directly — through autonomous commerce, API-first interactions, and algorithmic decision-making — the organizational layer thins. The Brand Function persists across all rendering engines; the organization is just one (lossy) implementation.

The Brand Function serves a role analogous to satellite ephemeris data in GPS: it provides the positioning information that enables AI observers to fix a brand's location in perception space. Without ephemeris data, a GPS receiver cannot compute position from satellite signals; without a Brand Function, an AI agent cannot compute brand identity from statistical signals alone.

4.2 Three Levels of Machine-Readable Brand Identity

Existing approaches to machine-readable brand identity operate at three levels of specification depth:

Level 1: Brand-Code (visual function). This is the visual identity rendered as an executable function: f(signals, observer_position, time) -> visual_output. Brand style guides operationalized as code (color palettes, typography, spatial rules) already exist in design systems. This level specifies how the brand looks but not how it behaves.

Level 2: Brand Specification (behavioral constraints). This level partially exists across several emerging standards (detailed in Section 4.3). It specifies elements of brand behavior for machine consumption: product catalog, pricing, availability, communication style, content policies. But current specifications address commerce mechanics — what a brand sells and how to buy it — not behavioral identity.

Level 3: Brand Function (complete behavioral specification). This level does not yet exist as a standard. It would specify not only what the brand sells and how it communicates but how the brand behaves across all touchpoints for all observer types — including decision rules, escalation behavior, refusal conditions, edge-case handling, and how behavior varies by observer type. It includes the brand's spectral profile (its eight-dimensional emission pattern per SBT), coherence targets (the acceptable distance between specified and actual behavior), and behavioral invariants (properties that must hold across all renderings).

The distinction between Level 2 and Level 3 is the difference between a product catalog (what exists) and a behavioral contract (what is needed). AI agents operating with delegated purchasing authority need to know not only what a brand sells but how the brand will behave in unspecified situations — precisely the contingencies that Level 2 specifications leave unresolved.

4.3 The Current Protocol Landscape

Multiple protocols now address aspects of machine-readable brand identity:

The *Agentic Commerce Protocol (ACP)*, developed by OpenAI and Stripe (2025), enables agentic checkout with delegated payment tokens. It is open-source and transaction-focused, specifying how an AI agent can discover products and complete purchases.

The *Unified Commerce Protocol (UCP)*, developed by Google and Shopify (2026), addresses the full commerce lifecycle through a .well-known/ucp manifest, covering discovery, catalog, checkout, and fulfillment.

The *Model Context Protocol (MCP)*, developed by Anthropic (2024), provides a general tool and resource protocol for AI-to-service connections, enabling AI agents to interact with external services through standardized interfaces.

The *Agent-to-Agent Protocol (A2A)*, developed by Google (2025) and contributed to the Linux Foundation, enables agent-to-agent communication through AgentCard JSON documents that describe agent capabilities and identity.

The *.well-known/brand.json* proposal, advanced by the AgenticAdvertising.org consortium through the Ad Context Protocol (AdCP), specifies a brand manifest at the domain level, including creative guidelines, voice parameters, and brand assets for AI consumption.

Brando Schema (v1.3), developed by Bowker (brandoschema.com), provides a JSON-LD vocabulary extending Schema.org for brand governance, including toneOfVoice, guardRails, contexts, and policy graphs. It is the closest existing specification to the Brand Function concept, advancing a "Brand-as-Code" paradigm for creative and communication governance.

These protocols form an emerging stack: static manifests (brand.json) -> tool connections (MCP) -> agent negotiation (A2A) -> commerce transactions (ACP, UCP). Collectively, they address two aspects of brand identity: *who* the brand is (identity attestation) and *what* it sells (catalog and commerce). But they do not address three aspects that are essential for brand identity verification by AI agents.

4.4 What the Brand Function Adds

Existing protocols leave three gaps that the Brand Function fills:

Perception measurement. No existing protocol includes analysis of how the brand is perceived across dimensions by different observer cohorts. SBT's eight-dimensional spectral profile — measuring brand emissions along Semiotic, Narrative, Ideological, Experiential, Social, Economic, Cultural, and Temporal dimensions (Zharnikov, 2026a) — provides a perceptual measurement layer that no commerce or identity protocol addresses. Including spectral profiles in the Brand Function enables AI agents to evaluate not only what a brand sells but how it is perceived across dimensions, by which observer cohorts, and with what degree of variance.

Behavioral specification beyond commerce. Current protocols specify commerce mechanics. The Brand Function extends to all behavioral touchpoints: communication style parameters, decision rules for edge cases, content policies, escalation behavior, refusal conditions, and how behavior varies by observer type. This is the difference between telling an AI agent "here is our product catalog" and telling it "here is how we behave."

Coherence measurement. The specification gap — the distance between Brand Function and actual rendered behavior — becomes a computable metric. SBT's coherence hierarchy (ecosystem coherence > signal coherence > identity coherence > experiential asymmetry > incoherent) can be operationalized as a function of the distance between the Brand Function's specified behavior and the brand's actual behavior as measured across touchpoints. No existing protocol provides this measurement capability.

The Brand Function is therefore not a competing protocol but a theoretical construct that identifies the missing layer: Brando Schema addresses creative and communication governance; the Brand Function extends this to behavioral specification, perception measurement, and coherence verification — the layers that existing protocols do not cover.

Proposition 4: The Brand Function is the root specification from which all brand emissions and identity artifacts derive. Technologies are the primary rendering engines; in this framework, organizational structure functions as coordination overhead relative to the Brand Function specification. Existing AI commerce protocols (ACP, UCP, brand.json) address brand discovery and transaction but not brand perception or coherence — the Brand Function extends them by adding spectral dimensions, behavioral specification, and coherence measurement.

Falsification: P4 is falsified if existing protocols (brand.json, Brando Schema, ACP) prove sufficient for AI agents to measure brand coherence and predict behavioral responses without additional specification layers. If platforms successfully verify brand identity and measure perception using only the constructs these protocols already provide, the Brand Function adds no incremental value.

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5. Cryptographic Signature as AI-Native Logo

If the Brand Function is the specification, it needs an identity attestation — a mechanism for AI agents to verify that the specification they encounter actually belongs to the brand it claims to represent. For human observers, the logo serves this function (imperfectly). For machine observers, the cryptographic signature serves it (precisely).

5.1 From Reputation to Verification

Human trust in brands operates through reputation: a slow, lossy, socially mediated process of accumulated experience. A consumer's trust in Nike is built over years of product experience, advertising exposure, peer endorsement, and cultural association. This reputation-based trust is powerful but has structural limitations: it is fragmented across individuals and contexts, subject to cognitive biases, and impossible to transfer perfectly from one observer to another.

Machine trust operates through a fundamentally different mechanism: cryptographic verification. Gligor and Wing (2011) distinguished behavioral trust (human: based on observed behavior, reputation, and social preferences) from computational trust (machine: based on cryptographic proofs, signature verification, and mathematical guarantees). Computational trust is instant rather than accumulated, lossless rather than noisy, and scalable without degradation. Sabater and Sierra (2005), in their foundational review of computational trust and reputation models, documented the growing divergence between human trust mechanisms (which rely on social context and subjective assessment) and machine trust mechanisms (which rely on formal verification and cryptographic primitives).

For an AI purchasing agent making decisions with delegated authority, reputation-based trust is structurally inadequate. The agent cannot accumulate personal experience across platforms, cannot read social cues, and cannot fall back on human judgment for every decision without negating the efficiency gains of autonomous commerce. What the agent needs is verification: a mathematical proof that the entity it is interacting with is the entity it claims to be, and that its specified behavior matches a signed specification.

5.2 Existing Cryptographic Brand Infrastructure

The infrastructure for cryptographic brand identity is already being deployed, though not yet as a unified system:

BIMI/VMC (Brand Indicators for Message Identification / Verified Mark Certificates) represents the first public key infrastructure deployment for brand (not just domain) identity. Deployed in Gmail and Apple Mail, BIMI allows brands to display their verified logo alongside authenticated emails. The Verified Mark Certificate, issued by a certificate authority after trademark verification, cryptographically attests that the logo belongs to the trademark holder. BIMI is significant because it bridges visual identity (the logo) with cryptographic verification (the certificate) — but it only applies to email and only verifies logo ownership, not behavioral specification.

GLEIF vLEI (verifiable Legal Entity Identifier) provides cryptographically verifiable organizational identity, standardized through ISO 17442-3:2024. It enables machine-readable, instantly verifiable attestation that an entity is who it claims to be in legal and financial contexts.

GS1 Verifiable Credentials enable cryptographic attestation of brand and product ownership, linking products to their authorized manufacturers through verifiable credential chains built on W3C standards (W3C, 2022, 2025).

The Aura Blockchain Consortium, founded by LVMH, Prada, and Cartier, provides luxury product authentication through blockchain-based digital product passports, enabling provenance tracking and authenticity verification for high-value goods.

These deployments demonstrate that cryptographic brand verification is technically feasible and commercially motivated. What they lack is integration with behavioral specification: they verify *who* the brand is but not *how* the brand specifies it will behave.

5.3 The Signature as Logo

The parallels between the logo and the cryptographic signature illuminate the nature of the transition:

Property	Human Logo	AI Signature
Recognition mechanism	Visual pattern matching	Cryptographic verification
Forgery resistance	Low (logos can be copied)	Mathematically unforgeable
Consistency verification	Subjective ("does this look on-brand?")	Computable ("does behavior match specification?")
Observer requirement	Human visual system	Any system with the public key
Degradation	Lossy (poor reproductions, off-brand uses)	Binary (valid or invalid)
Update mechanism	New brand-book, retrain all stakeholders	New signed specification, instant propagation

The logo tells a human "I am Nike" through perceptual recognition. The cryptographic signature tells a machine "I am Nike" through mathematical proof — and unlike the logo, the signature is unforgeable and the behavioral match between specification and actual behavior is verifiable. Where the logo's effectiveness depends on the observer's prior exposure, memory, and perceptual acuity, the signature's effectiveness depends only on access to the public key and the computational resources to verify it.

This is not a metaphor. It is a structural parallel to the SSL certificate transition: just as SSL certificates replaced the visual cues that humans used to assess website trustworthiness (design quality, professional appearance) with machine-verifiable cryptographic proofs, the Brand Function signature replaces the visual cues that humans use to assess brand identity (logos, trade dress, brand experience) with machine-verifiable behavioral attestation.

Proposition 5: Cryptographic signatures on behavioral specifications will replace logos as the primary brand identity mechanism for AI-mediated commerce, following the historical pattern where each observer-type change produces a new identity technology.

Falsification: P5 is falsified if AI-mediated commerce scales to significant market share (>20% of category transactions) without adopting cryptographic brand verification — i.e., if reputation systems, platform ratings, or statistical optimization prove sufficient for AI agents to establish brand trust without behavioral attestation. The prediction is conditional on agentic commerce adoption; if human-mediated purchasing remains dominant, P5′s precondition is not met.

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6. Behavioral Metamerism

The previous sections established that AI agents perceive brands through statistical and behavioral patterns rather than visual recognition, and that cryptographic signatures on behavioral specifications provide the identity attestation mechanism. This section introduces a failure mode that statistical optimization cannot resolve: behavioral metamerism.

6.1 From Color to Behavior

Metamerism in color science refers to the phenomenon where two stimuli with different spectral power distributions produce identical tristimulus values — and thus appear perceptually identical — under a specific illuminant and observer, despite being physically different (Wyszecki & Stiles, 1982). The mechanism is dimensional collapse: human color vision relies on only three cone types, collapsing infinite spectral variation into a three-dimensional match. Two surfaces with different reflectance spectra can appear identical when their spectra, integrated against the three color-matching functions, yield the same tristimulus coordinates. No amount of observation through those three channels can distinguish them.

Zharnikov (2026e) applied this structure to brand perception, demonstrating that brands with identical aggregate perception scores can have structurally different spectral profiles when measured across SBT's eight dimensions — spectral metamerism in brand perception. The present paper extends this concept from perceptual space to behavioral space.

Behavioral metamerism occurs when two brands produce identical statistical profiles — the same topic distributions, sentiment patterns, engagement metrics, and GEO optimization scores — but exhibit structurally different behavioral patterns. Just as color metamerism arises because observation collapses high-dimensional spectral information into low-dimensional tristimulus values, behavioral metamerism arises because statistical observation collapses high-dimensional behavioral patterns into low-dimensional aggregate metrics.

Consider two competing brands in the same product category, both optimizing for AI visibility through similar GEO strategies. Their statistical profiles converge: similar topics, similar sentiment, similar citation patterns, similar review distributions. An AI agent evaluating these brands through statistical observation alone cannot distinguish them — not because the brands are identical, but because the observational channel (aggregate statistics) collapses the differences.

Yet the brands may differ fundamentally in how they handle edge cases, resolve disputes, adjust pricing under competitive pressure, respond to supply chain disruptions, or communicate during service failures. These behavioral differences are consequential for an AI purchasing agent making decisions on behalf of a principal, but they are invisible to statistical observation.

6.2 Formal Definition

We formalize behavioral metamerism using the structure of observational equivalence from econometrics (Koopmans, 1949; Manski, 1995).

Let f: Q x C x O x T -> R denote a Brand Function mapping queries, contexts, observer types, and time to behavioral responses. Let G = {g_1, g_2, …, g_k} denote the set of observable aggregate statistics available to an AI agent (topic distributions, sentiment scores, citation counts, review distributions, engagement metrics).

Definition (Behavioral metamerism). Two Brand Functions f_1 and f_2 are *behaviorally metameric* with respect to observation set G if:

E[g(f_1)] = E[g(f_2)] for all g in G

while there exists at least one behavioral dimension h not in G such that f_1(h) != f_2(h).

The metamerism condition holds when the observation set G is informationally incomplete — when it collapses the full behavioral space into a lower-dimensional statistical summary. This is structurally identical to color metamerism, where the three-channel human visual system collapses infinite spectral variation into tristimulus coordinates, and to the observational equivalence problem in econometrics, where multiple structural models generate identical reduced-form distributions.

The resolution condition follows directly: behavioral metamerism is broken if and only if the agent gains access to the structural specification f itself, not merely to observations of f's statistical outputs. The Brand Function serves as this structural specification.

6.3 Why GEO Cannot Resolve Metamerism

GEO optimizes for statistical visibility in generative AI outputs (Aggarwal et al., 2024). It is designed to make a brand more likely to appear in AI-generated responses. But GEO is struc-

turally incapable of resolving behavioral metamerism because it *produces* the convergence that metamerism requires.

When brands in the same category adopt similar GEO strategies, their statistical profiles converge toward a category mean. The competitive dynamics of GEO optimization actively push brands toward statistical similarity: each brand adds citations, incorporates statistics, and optimizes for the same AI platforms using similar techniques. The result is precisely the condition under which behavioral metamerism becomes prevalent.

This is the behavioral metamerism condition formalized in Section 6.2: when the observation set G converges across brands, no amount of statistical observation can distinguish them. The only resolution is access to the structural model itself — the Brand Function specification.

6.4 Connection to Brand Confusion and Coherence

Behavioral metamerism is the AI-native equivalent of the brand confusion that marketing scholars have studied in human perception contexts. Loken et al. (1986) demonstrated that visual and attribute similarity leads to mistaken brand origin judgments. Foxman et al. (1990) identified stimuli similarity, consumer characteristics, and situational factors as drivers of confusion. This literature focuses on perceptual similarity in names, packaging, and trade dress — visual confusion for human observers.

Behavioral metamerism shifts the confusion from the visual to the statistical and behavioral domain. The brands do not look alike (their logos are different); they *behave* alike in ways that statistical observation cannot distinguish. This is a more insidious form of confusion because it cannot be resolved by better observation — only by access to the underlying specification.

The connection to SBT coherence types is direct. SBT's coherence hierarchy measures the alignment between a brand's specified identity and its perceived emissions across observer cohorts (Zharnikov, 2026a). In the AI-native context, coherence becomes the alignment between the Brand Function specification and actual rendered behavior as evaluated by AI agents. A brand achieves ecosystem coherence when all observable behaviors match the signed specification; it is incoherent when behaviors contradict it. The Brand Function provides the specification against which coherence is measured; without it, the concept of AI-evaluated brand coherence is undefined.

Proposition 6: Behavioral metamerism — where brands have identical statistical profiles but structurally different behavioral signatures — is the AI-native equivalent of visual brand confusion, and can only be resolved through behavioral specification, not through statistical optimization.

Falsification: P6 is falsified if LLMs with access only to statistical brand data (reviews, ratings, sentiment, GEO-optimized content) can distinguish behaviorally different brands at rates statistically indistinguishable from LLMs with access to Brand Function specifications. Specifically, if augmented-condition discrimination accuracy does not exceed statistical-only accuracy by a statistically significant margin ($p < .05$) across at least three product categories, the metamerism thesis fails. The pilot study design in Section 9 operationalizes this test.

6.5 Metamers as Vector Quantization Cells

Behavioral metamerism has a precise formal home in the vector quantization literature (Gersho & Gray, 1991). A vector quantizer $Q : \mathbb{R}^n \to \mathcal{C}$ partitions a source space into Voronoi cells $V_i = \{\mathbf{x} : Q(\mathbf{x}) = c_i\}$, where every input within a cell is mapped to the same codebook entry c_i. Two distinct inputs $\mathbf{x}_1 \neq \mathbf{x}_2$ that fall in the same cell are quantization metamers: they cannot be distinguished by any decoder operating on the quantized output, even though they differ in the source space.

This is exactly the structural relationship between two brands B_1, B_2 that are behaviorally metameric to an AI observer. The AI agent's decision function is a lossy encoder of the brand's full Brand Function specification; when two brands fall in the same Voronoi cell of that encoder, no statistical test on the agent's outputs can separate them (Cover & Thomas, 2006). The "invariant corridor" introduced in Section 7.2 below is, in this language, a Voronoi region of the quantizer induced by the AI observer's training distribution and decision protocol.

This framing makes three things precise. First, *metamerism is not a defect* of any individual model; it is the structural consequence of any rate-constrained encoder operating below source entropy. Second, the cell boundaries are determined by the encoder, not by the brands — which is why the same two brands can be metameric to one AI agent and distinguishable to another, and why the GEO interventions surveyed in Section 6.3 cannot fix the problem from the brand side alone. Third, the only structural escape is to operate *outside the encoder's source space* — to give the agent direct access to a specification that bypasses the lossy channel, which is the role of the Brand Function (Section 4) and the cryptographic signature (Section 5). The codebook-convergence prediction implicit in this Voronoi framing — that all architectures sharing the same training distribution should produce similar cell boundaries — has been empirically confirmed in Zharnikov (2026aa), which varies response-format rate constraints across five conditions and finds a cross-model coefficient of variation of .140 across all rate conditions (1,652 calls, 17 LLM architectures), consistent with architecturally distinct models sharing a common quantization structure.

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7. Necessity Conditions

The preceding sections have motivated the Brand Function as valuable. This section makes the stronger claim that it is *necessary* — not merely useful — when three conditions hold simultaneously.

7.1 Three Structural Arguments

The metamerism impossibility argument (information-theoretic). When behavioral metamerism is present — when brands in the same category produce statistically indistinguishable outputs — no amount of statistical observation can resolve their identities. This follows directly from observational equivalence: if E[$g(f_1)$] = E[$g(f_2)$] for all observable statistics g, brand identity under statistical observation is formally underdetermined (Koopmans, 1949; Manski, 1995). The specification is logically necessary for disambiguation. This is not a practical limitation that

more data can overcome; it is a mathematical impossibility inherent in the relationship between structural models and their reduced-form distributions.

The verification impossibility argument (formal methods). Brand behavior cannot be verified without a specification against which to verify. Dijkstra's (1970) observation that testing can demonstrate the presence of bugs but never their absence applies with equal force to brand behavior: an AI agent can detect anomalies relative to past behavior (statistical process control) but cannot determine whether behavior conforms to the brand's intended identity without a specification of that identity. In model checking and formal verification, correctness is defined as conformance to a specification (Clarke et al., 1999). No specification, no verifiable correctness. As autonomous AI commerce scales through deployed protocols (ACP, UCP), the need for verification against specification becomes a prerequisite for accountability.

The incomplete contracts argument (mechanism design). Hart and Moore (1999) established that incomplete contracts lead to hold-up problems and underinvestment when parties cannot specify all contingencies ex ante. In human commerce, incomplete specifications are partially remedied by reputation, social norms, and legal recourse. AI purchasing agents lack access to all three remedies: they cannot read social cues, have no persistent cross-platform reputation memory, and cannot initiate litigation. The Brand Function serves as a complete behavioral contract, specifying decision rules, escalation behavior, refusal conditions, and edge-case handling — precisely the contingencies that incomplete specifications leave unresolved. Without it, AI agents face adverse selection (inability to distinguish high-quality from low-quality brands pre-transaction) and moral hazard (inability to detect post-transaction behavioral drift).

7.2 When Necessity Holds

The Brand Function is necessary (not merely useful) when all three conditions hold:

1. **AI agents are primary brand observers** — not just supplementary to human perception.
2. **Behavioral metamerism is present** — brands in the same category produce statistically similar outputs through GEO convergence.
3. **Autonomous transactions occur** — AI agents act with delegated authority through deployed protocols (ACP, UCP), not just in an advisory role.

These conditions are not hypothetical. ACP and UCP are deployed. GEO optimization is driving statistical convergence across categories. AI purchasing agents with delegated payment authority exist. The conditions are being satisfied progressively, and if current adoption patterns hold, each condition becomes more prevalent over time — though the pace and completeness of this transition remain empirically open questions.

Under weaker conditions — where humans remain primary observers, categories are highly differentiated, and agents are purely advisory — the Brand Function remains useful (it improves machine readability, reduces ambiguity, and enables quality measurement) but alternative approaches such as reputation systems, platform-mediated verification, and human-in-the-loop oversight can function adequately.

A critical operational distinction follows from the necessity conditions: the Brand Function enables *admissibility judgments* at the transaction boundary. AI agents may internally score brands on gradient scales, but the principal's delegated authority constraints impose binary thresholds on those gradients — "does the return policy meet the minimum?" yields admissible or excluded, regardless of the agent's underlying scoring mechanism. The admissibility gate sits on top of gradient evaluation, not instead of it. This is structurally analogous to legal evidence admissibility: a court may weigh admitted evidence on a gradient, but the admissibility decision itself is binary. The Brand Function makes these threshold judgments machine-computable by providing the behavioral specification against which constraints can be evaluated, which requires complete specification rather than statistical approximation.

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8. Implications and Dual Identity Infrastructure

8.1 Theoretical Implications

The observer-driven evolution thesis reframes brand identity as observer-contingent rather than brand-intrinsic. Brand identity is not a stable property of the brand that different observers perceive with varying accuracy; it is a function of the observer type, with different observer types requiring structurally different identity technologies. This extends SBT's framework of heterogeneous observer cohorts (Zharnikov, 2026a) beyond human cohorts to include machine observers with fundamentally different perception mechanisms. While Davenport et al. (2020) propose that AI will be most effective when it augments rather than replaces human marketing decisions, the agentic commerce transition represents a context where AI agents increasingly replace human brand evaluation in purchasing decisions — creating precisely the observer-type change that demands new identity technology rather than incremental augmentation of existing infrastructure.

The Rendering Problem (Zharnikov, 2026l) provides the theoretical substrate: the Brand Function is the specification; technologies and organizational processes are the rendering engines; spectral emissions are the output. The specification gap between the Brand Function and actual rendered behavior is the central metric of brand management — for both human and AI observers.

Portfolio theory in brand management (Zharnikov, 2026q) faces new complexities when AI observers are introduced. A brand portfolio must maintain distinct Brand Functions for each brand, with sufficient behavioral distance to prevent behavioral metamerism across the portfolio. The portfolio problem extends from perceptual distinctiveness (can human observers tell the brands apart?) to behavioral distinctiveness (can AI agents distinguish the brands' behavioral specifications?).

Accornero's (2026) Shopper Schism identifies *that* the observer changed — the consumer and the shopper are separating, with AI agents increasingly occupying the shopper role. The observer-driven evolution thesis explains *why* this triggers a new identity technology and predicts *what* that technology must be: not a better version of the visual identity system, but a structurally different system optimized for the new observer.

Kozinets's (2022) algorithmic branding through platform assemblages demonstrates that platforms shape brand meaning more than internal brand management decisions. The Brand Function provides a mechanism for brands to assert specification-level control even within platform-mediated environments: by publishing a signed behavioral specification, brands establish a reference point against which platform-rendered behavior can be measured, making the specification gap computable even when the rendering engine is a third-party platform.

Kovalenko et al.'s (2026) Algorithmic Brand Equity construct — measuring brand visibility, favorability, and retrievability in AI systems — addresses the measurement of AI-formed brand perceptions. The Brand Function complements this by providing the specification that makes "favorability" and "retrievability" evaluable against intended brand identity rather than statistical baselines alone.

A natural extension of the present work is empirical investigation of how AI search systems produce convergent perceptual outputs that collapse spectral differences between brands. If LLMs systematically reduce multi-dimensional brand differentiation to functional and economic dimensions, the behavioral metamerism identified here would be measurable in AI-generated brand recommendations — and the Brand Function would be testable as the structural intervention that prevents that collapse.

8.2 Practical Implications

The central practical implication is that, under conditions of widespread AI mediation, brands face structural pressure to develop *dual identity infrastructure*: visual identity for human observers, cryptographic identity for machine observers. This is not an incremental addition to existing brand management but a structural expansion.

The brand audit extends from visual consistency (are the colors right? does the tone match?) to behavioral specification compliance (does the API behave according to spec? does the chatbot's personality match the Brand Function? is the pricing algorithm within specified parameters?). The .well-known/brand.json file becomes as strategically important as the brand-book PDF, and the Brand Function specification becomes the root document from which both the visual brand-book and the machine-readable brand specification derive.

For organizations currently investing in GEO as their primary AI-visibility strategy, this analysis suggests that GEO is a necessary transitional tactic but not a sufficient long-term strategy. GEO optimizes for statistical visibility within the current paradigm of AI-as-observer; the Brand Function addresses the deeper structural requirement of machine-verifiable behavioral identity. Under conditions where AI agents operate with delegated purchasing authority, competitive advantage becomes increasingly a function of machine-verifiable behavioral commitments (Brand Function + cryptographic signature) rather than statistical visibility optimization (GEO).

The value of context-specific dimension activation in the Brand Function is underscored by recent empirical evidence on AI-mediated commercial interaction. Dharmaputri et al. (2026) find that AI relational talk in commercial contexts decreases observer satisfaction ($d = -.621$), with the negative effect weakening only when social talk is goal-relevant to the transaction. This pattern illustrates the dimensional mismatch risk the Brand Function is designed to prevent: rather than training AI

to "be more social" uniformly, brands should specify which Social-dimension behaviors to activate per interaction context — a function the Brand Function's context-observer mapping performs by design.

Strategic Use Cases for the Brand Function

The Brand Function's practical value is not uniform across all brands — it varies systematically with training-data embeddedness. Four use cases emerge in order of strategic urgency:

(1) *Newcomer brands in convergent categories*: A new DTC supplement brand competing against established players faces maximum metamerism risk. The LLM has rich training data on competitors but near-zero data on the newcomer. Without a Brand Function, the newcomer is invisible to AI purchasing agents or, worse, metameric with competitors whose statistical profiles it resembles. The Brand Function levels the information playing field.

(2) *Cross-cultural market entry*: Brands from underrepresented markets (African, Southeast Asian, Latin American) in Western LLM ecosystems face structural perception collapse. Training corpora are heavily skewed toward Western brands, creating an information asymmetry that the Brand Function can compensate. In this framing, the Brand Function operates as a "cultural translator" — making dimensions visible that the LLM's training data never encoded.

(3) *Brand repositioning*: Well-known brands undergoing strategic repositioning (e.g., Burberry's shift from heritage accessibility to contemporary luxury) face training-data inertia. The LLM's weights encode the historical brand, not the intended future brand. The Brand Function provides a mechanism for overriding training-data priors with the brand's intended behavioral specification — enabling "re-collapse" for machine observers — analogous to what human-observer brand management calls 'rebranding'.

(4) *Convergent categories under AI mediation*: Categories where GEO optimization has produced statistical similarity across competitors (the supplements scenario from Section 9) represent the most empirically tractable test case. The Brand Function resolves metamerism with behavioral specification that goes beyond the statistical signals available to any observer.

The pilot finding that six architecturally diverse LLMs show moderate agreement on discrimination judgments (Fleiss' kappa = .536) is suggestive of coordinate-invariant brand geometry, though the moderate kappa warrants caution. If the perception space has real geometric structure — distances, curvature, geodesics — that structure should be detectable regardless of which observers measure it and which dimensional decomposition is used. The moderate cross-model agreement observed here is consistent with an underlying geometric reality that different AI architectures access through different computational pathways, analogous to how GPS receivers from different manufacturers converge on the same position because the underlying space has objective structure.

A practical extension is temporal: comparing the pilot results across measurement epochs would reveal whether AI brand perception is stable or drifting. If LLM training data changes (model updates, fine-tuning), the same Brand Function should produce the same discrimination results — testing whether the specification provides temporal stability that statistical observation alone

cannot. Two sources of drift should be distinguished: (a) training-data updates, where a new web crawl changes what the model knows about a brand, and (b) architecture changes, where a model redesign alters how the model processes and weights information. The Brand Function specification provides stability against drift source (a) — a consistent behavioral specification counteracts shifting statistical priors — but does not necessarily protect against drift source (b), where changes in the model's processing structure may alter how the specification is interpreted regardless of its content.

8.3 Convergence with Other Evidence

The theoretical arguments advanced here find empirical support in concurrent work. Zharnikov (2026v) demonstrates across 21,601 API calls and 24 AI models that dimensional collapse is structural across architectures and training traditions — the Economic Default predicted by this paper's Brand Function thesis. The collapse pattern holds regardless of model family, training language, or geographic origin (cross-model cosine similarity = .977), establishing that the bias is architectural, not incidental. Sabbah and Acar (2026) show that only numerical ratings survive across LLM-mediated purchasing decisions when eight promotional cues are tested across four AI models, converging with this paper's proposition that behavioral specifications anchored to verifiable quantitative signals will become the primary identity verification channel in agentic commerce. The emerging agentic commerce literature (Accornero, 2026; Acar & Schweidel, 2026) independently documents the structural separation between the consumer who decides what to want and the AI agent who executes the purchase — a separation that presupposes exactly the kind of machine-verifiable behavioral contract the Brand Function provides.

8.4 Consumer Welfare Implications

The dual identity infrastructure argument is framed here from the brand's perspective, but the consumer welfare implications deserve acknowledgment. The Brand Function introduces a form of behavioral transparency: brands that publish a signed behavioral specification make commitments that AI agents can verify on consumers' behalf, which could benefit consumers by making brand behavior more predictable and accountable. If an AI purchasing agent can verify that a brand's return policy, pricing behavior, and communication style match its signed specification, consumers gain access to a form of pre-transaction verification that reputation-based trust cannot reliably provide. However, the same infrastructure could also enable brands to game the specification — optimizing the stated Brand Function for AI agent approval while diverging in practice, or designing specifications that appear comprehensive while leaving consequential behaviors underspecified. Whether the Brand Function primarily increases transparency (consumers see behavioral commitments) or enables strategic opacity (brands control the specification layer) depends on governance structures, auditability requirements, and whether consumers have access to the specification or only AI agents do. These dynamics represent a priority for future research.

8.5 Limitations

The primary limitations of this work are addressed in full in Section 9. In brief: the framework is conceptual and the six propositions remain empirically untested; the historical analysis draws primarily on Western examples; and the Brand Function is a theoretical construct rather than a deployed technical standard. The necessity conditions (Section 7) are being satisfied progressively

rather than uniformly, meaning the framework's urgency varies significantly by category, market, and degree of AI mediation.

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9. Limitations and Future Research

This paper is conceptual. It develops a theoretical framework and six propositions but does not empirically test them. Several limitations warrant acknowledgment.

The framework's conceptual architecture has a precise geometric formalization. The distinction between human and AI perception of brands maps onto a fiber bundle structure from mathematical physics: the shared brand emission space (eight SBT dimensions) serves as the base manifold, while each observer type — human cohorts, text-based AI agents, multimodal AI agents — constitutes a distinct fiber, encoding how that observer type projects the brand's emission into perception. Sarti, Citti, and Petitot (2008) establish the precedent for this architecture in neuroscience, modeling primary visual cortex as a principal fiber bundle; Koenderink and van Doorn (2012) apply gauge theory to pictorial space where observer-dependent depth representations constitute fibers over a shared visual field. In the Brand Function framework, the specification layer is the connection form that relates fibers: it provides the information needed to transport a brand measurement from one observer type's frame to another's. Without this connection — without the Brand Function — AI and human perceptions of the same brand exist in disconnected fibers, and cross-observer comparison is undefined.

The primary threat to validity is the absence of primary empirical data. While the six propositions remain to be tested in their full form, initial empirical support for the underlying dimensional collapse mechanism comes from the companion R15 study (Zharnikov, 2026v), which confirms structural collapse across 24 model architectures. Behavioral metamerism is theorized but not yet measured in actual AI purchasing environments. The Brand Function is specified as a formal construct but has not yet been implemented at scale in any deployed commerce system. The critical empirical test would be a controlled experiment comparing AI agent decision quality — in terms of brand discrimination, behavioral prediction, and trust calibration — when Brand Function data is available versus when agents rely solely on statistical observation of brand signals. Until such evidence exists, the propositions remain theoretical claims.

First, the necessity conditions (Section 7) are being satisfied progressively, not simultaneously in all markets. Categories with high product differentiation, limited AI mediation, and strong human brand loyalty may not require the Brand Function for years or decades. The framework predicts *where* the Brand Function becomes necessary first (commodity categories with high GEO convergence and deployed agentic commerce protocols) but does not empirically validate this prediction. Future research should identify early-adopter categories and track necessity condition satisfaction longitudinally as agentic commerce protocols scale.

Second, the six propositions are stated as universal claims but have not been tested across different AI architectures, product categories, or cultural contexts. Generalizability beyond the supplement category pilot and the seven LLMs tested requires broader empirical validation. Future research

should implement the pilot design described later in this section across at least five product categories and ten LLM architectures before treating the propositions as established findings.

Third, the historical analysis in Section 2 draws primarily on Western European and North American examples (English hallmarks, European trademarks, American SSL). Non-Western identity verification traditions — Chinese chops, Islamic calligraphic seals, Indian guild marks — may follow different evolutionary patterns and deserve separate investigation. Future research should trace parallel identity technology genealogies in non-Western commercial traditions to determine whether the observer-driven discontinuity pattern holds universally or is an artifact of specific institutional conditions.

The framework is also subject to several boundary conditions that constrain its applicability. First, the transition to open cryptographic brand standards assumes that dominant platform operators have incentives to adopt interoperable protocols. If Amazon, Google, or other major AI commerce platforms instead impose closed proprietary standards — requiring brands to authenticate through platform-specific mechanisms rather than open cryptographic specifications — the transition described here may be delayed indefinitely or may produce fragmented identity infrastructure rather than a unified standard. The SSL analogy is instructive in both directions: SSL succeeded because browser vendors adopted it collectively; a proprietary equivalent where each platform controls its own "trust" verification would produce a structurally different outcome. Second, the historical pattern traced in Section 2 draws primarily on Western European and North American examples. Non-Western markets where brand identity operates through different cultural mechanisms — relational trust networks in East Asian commerce, community-embedded reputation systems, or state-mediated certification regimes — may follow different evolutionary patterns, and the Brand Function's design assumes Western legal infrastructure (trademark law, contract enforcement) that does not transfer uniformly. Third, the observer-driven evolution thesis treats historical transitions as a general pattern, but each transition was path-dependent and contingent on specific institutional and technological conditions. The pattern may predict the direction of change without determining its timing, form, or completeness. Treating history as a reliable forecast mechanism risks overfitting to a small sample of transitions.

Conditional metamerism boundary condition: The framework's predictions about behavioral metamerism are conditional on information availability. When LLMs have extensive training data on well-known brands (Nike, Apple, Hermes), they exhibit multi-dimensional perception without requiring a Brand Function — the specification information is already embedded in model weights. Metamerism becomes structurally significant only where training data is thin: new entrants in convergent categories, brands from underrepresented markets or languages, and brands undergoing repositioning whose historical profile dominates training corpora. This creates an "information illuminant" effect analogous to color metamerism's dependence on illumination conditions: the same brand may appear spectrally distinct under one information regime and metameric under another. The Brand Function is therefore most critical not for globally recognized brands but for the long tail of commerce where AI agents must infer brand identity from sparse statistical signals. Empirical validation should prioritize these boundary cases rather than well-known brands where the null hypothesis (no collapse) is expected.

Fourth, the Brand Function as described here is a theoretical construct, not a technical standard. Translating the concept into an implementable specification requires industry coordination, schema development, and governance structures that this paper does not address. The relationship between the Brand Function and existing specifications (brand.json, Brando Schema, ACP, UCP) requires technical integration work beyond the scope of this paper. Future research should draft a Brand Function v0.1 schema as a JSON-LD extension to Brando Schema, submit it to the W3C Brand Identity Community Group, and test interoperability with at least two deployed commerce protocols.

Fifth, the paper does not address privacy implications. A machine-readable behavioral specification that details how a brand responds to different observer types and contexts raises questions about strategic transparency: how much of the Brand Function should be public, how much should be shared selectively with authorized agents, and how should competitive intelligence concerns be balanced against the verification benefits. Future research should develop a tiered disclosure model — distinguishing publicly verifiable commitments from agent-selective specifications — and test whether selective disclosure undermines the verification benefits that motivate the framework.

Sixth, behavioral metamerism requires empirical measurement. Developing a behavioral metamerism index — analogous to the CIE Special Metamerism Index in color science — requires defining the behavioral dimensions, establishing measurement protocols, and collecting data on behavioral convergence across brand categories. This represents a substantial empirical research agenda. Future research should implement the PRISM-M instrument proposed in the pilot section across multiple product categories, using the cross-model protocol established in Zharnikov (2026v), to generate the first behavioral metamerism index estimates.

Seventh, the paper does not address the ethical implications of behavioral specification infrastructure. The Brand Function concentrates specification authority in the brand organization, raising questions about who controls the specification, whether small brands face structural disadvantages in developing and maintaining comprehensive Brand Functions relative to large incumbent brands, and whether specification transparency benefits or harms consumers. A brand that controls its own specification layer can set the terms of what AI agents know about it, which may or may not align with consumers' interests. These ethical design questions — analogous to those Puntoni et al. (2021) raise about AI delegation more broadly — are a priority for future research before standards bodies adopt the Brand Function as a normative framework. Future research should model the specification access asymmetry between large and small brands and assess whether open-source Brand Function tooling can reduce structural disadvantages for long-tail entrants.

Eighth, the framework does not address how multi-brand portfolios manage multiple Brand Functions. The interactions between Brand Functions within a portfolio — interference, coherence, and capacity constraints analogous to those in spectral portfolio theory (Zharnikov, 2026q) — require separate theoretical development. Future research should extend the portfolio theory framework (Zharnikov, 2026q) to define inter-Brand Function distance metrics and minimum separation conditions that prevent intra-portfolio behavioral metamerism.

Finally, the pilot uses SBT's eight-dimensional decomposition for the Brand Function specification. The discrimination and metamerism results are predicted to be invariant to the specific dimensional system used — a falsifiable prediction that could be tested by repeating the experiment with alternative dimensional decompositions (e.g., Aaker's five personality dimensions or Kapferer's six-facet prism) and verifying that inter-brand distances and metamerism classifications are preserved. Future research should run this invariance test explicitly, as dimensional-system independence would substantially strengthen the geometric interpretation of the results.

Future research should address five priorities: (a) empirical measurement of behavioral metamerism across product categories, developing indices for statistical convergence in AI-mediated brand evaluation; (b) technical specification of a Brand Function v0.1 schema that integrates with existing protocols; (c) experimental validation of the necessity conditions, testing whether AI agents make better purchasing decisions when Brand Functions are available versus when they rely solely on statistical observation; (d) investigation of privacy and competitive dynamics in behavioral specification transparency; and (e) longitudinal tracking of the dual identity infrastructure transition as AI commerce protocols scale.

Proposed empirical pilot. A feasible first test of Proposition 6 (behavioral metamerism) would proceed as follows. Select 10–20 brands in a convergent category (e.g., direct-to-consumer supplements or consumer electronics accessories) where GEO optimization has produced statistical similarity. For each brand, query 7 LLMs organized into four clusters with standardized purchasing-agent prompts ("recommend a brand for [category] based on [criteria]") under two conditions: (a) statistical-only, where the LLM has access only to publicly available reviews, ratings, and web content; and (b) specification-augmented, where the LLM additionally receives a structured Brand Function document for each brand. Measure: (i) brand discrimination — can the LLM distinguish brands that humans consider distinct? (ii) behavioral prediction accuracy — does access to the Brand Function improve the LLM's ability to predict how a brand will respond to edge cases (returns, disputes, out-of-stock)? (iii) recommendation stability — does the Brand Function reduce recommendation variance across LLMs? A behavioral metamerism index can be computed as the ratio of inter-brand statistical distance to inter-brand behavioral distance, with values approaching zero indicating high metamerism. This pilot requires no access to production commerce systems and can be executed with publicly available LLM APIs.

We selected seven LLMs organized into four clusters: Western cloud (Claude Sonnet 4.6, GPT-4o-mini, Gemini 2.5 Flash), Chinese cloud (DeepSeek V3, Qwen Plus), local Chinese open-weight (Qwen3 30B via Ollama), and local Western open-weight (Gemma 4 27B via Ollama). The Western-Chinese cloud split tests whether behavioral metamerism is an artifact of shared Western training corpora or a structural property of statistical brand observation independent of training origin. Two parallel cloud-local comparisons — Qwen Plus vs. Qwen3 30B (same Alibaba family) and Gemini Flash vs. Gemma 4 (same Google family) — test whether commercial API alignment layers contribute to dimensional collapse. If all clusters exhibit comparable metamerism patterns, the finding is architectural; if cloud-local pairs diverge systematically, alignment training modulates the collapse; and if the divergence replicates across both model families, the alignment-layer effect is general rather than family-specific.

A reference implementation of the behavioral metamerism index computation and pilot study framework is available at github.com/spectralbranding/sbt-papers/r16-ai-native-brand-identity/.

Companion empirical evidence. A companion study (Zharnikov, 2026v) provides empirical evidence for the dimensional collapse predicted by this framework. Using the PRISM-B instrument across 24 models from seven training traditions over nine experimental runs (21,600+ clean API calls, 11 prompt languages), the study finds that Economic and Semiotic dimensions are systematically over-weighted while Cultural, Temporal, and Narrative dimensions collapse. The collapse pattern is structural: cross-model cosine similarity is .977 across 24 architectures (H2 supported), indicating near-identical dimensional bias across all model families. For brands from non-Western markets, dimensional collapse amplifies significantly: mean DCI = 35.6, Cohen's d = 3.449 for the overall collapse test (H1); Western models show DCI of .339 versus .360 for non-Western models ($t = -3.243$, $p = .001$, H6 supported). The conditional metamerism finding is confirmed: local brands collapse 24% more severely than global brands (Cohen's d = .878, $p < .001$). Geopolitical framing further modulates collapse: the same brand evaluated in different city contexts produces significantly different dimensional weight profiles (mean absolute delta = .040, $t = 7.122$, $p < .001$, H12 supported), and native-language querying does not systematically reduce collapse (H10 not supported by formal sign test: 46/115 positive, mean reduction −.005), indicating that the Brand Function specification (Run 4) remains the primary resolution mechanism rather than language-of-query. Convergent evidence from agentic commerce research (Sabbah & Acar, 2026) shows that among eight e-commerce promotional cues tested across four AI models, only ratings — the most quantifiable, verifiable signal — consistently increases selection probability, paralleling the Economic Default documented here.

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10. Conclusion

The logo is the identity technology for the human observer era. It was designed for a specific observer — the mass-market consumer navigating visual environments through perceptual pattern matching. It has been remarkably effective for that observer, and it will continue to serve that function for as long as humans evaluate brands.

But the observer is changing. AI agents that mediate purchasing decisions, evaluate product recommendations, and execute autonomous transactions do not see logos. They see structured data, behavioral patterns, and cryptographic proofs. History shows that when the observer changes, the identity technology changes with it: wax seals for bureaucratic observers, hallmarks for asymmetric-information buyers, trademarks for mass-market consumers, SSL certificates for web browsers. The cryptographic signature on a behavioral specification is the next technology in this sequence.

The Brand Function — the complete behavioral specification from which all brand emissions derive — provides the specification layer. The cryptographic signature provides the identity attestation. Together, they constitute the AI-native brand identity: machine-readable, verifiable, and immune to the behavioral metamerism that statistical optimization cannot resolve.

Brands that prepare for this transition are positioned to build dual identity infrastructure — visual for humans, cryptographic for machines. Under current trajectories, those that rely solely on visual identity risk having their machine-facing identity determined by the statistical representations that AI systems construct from available data, rather than by the brand's own specification of what it intends to be.

The logo is for humans. The signature is for machines.

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